

CBE 30235: Introduction to Nuclear Engineering

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Course Description

Electrical energy generation is a huge and growing market as the world attempts to reduce reliance on fossil fuels. Nuclear power is a carbon-free way of producing the heat required to generate electricity. In this course, we shall examine the key facets of this generation method: how a sustained fission chain reaction can be safely achieved, how the heat generated can be safely removed, and how the radioactive waste products can be safely disposed of. The course introduces fundamentals of fission, including neutron capture cross-sections, reactor poisoning, cladding, and shielding. While focus is maintained on conventional Pressurized Water Reactors (PWRs), we will explore alternative approaches such as liquid metal and high-temperature gas reactors, Small Modular Reactors (SMRs), and the historical Light Water Breeder Reactor (LWBR) program.

Course Goals

- Review the fundamentals of nuclear physics with application to radioactive decay.
- Understand the physical principles underlying nuclear fission and reactor operation.
- Explore various reactor types, with an emphasis on PWRs and breeding cycles.
- Analyze thermal-hydraulic and safety considerations in reactor design.
- Review the future landscape of nuclear energy, including SMRs and waste disposal.

Required Text

Lamarsh, J.R. & Baratta, A.J. *Introduction to Nuclear Engineering*, 4th Ed., 2017.

Grading and Assessment

- **Daily Quizzes:** Conceptual/technical checks following each lecture.
- **Weekly Homework:** Quantitative problem sets illustrating key concepts.
- **Final Project (Design Review):** Group presentation and written report.
 - Class size $\approx$ 18 students (6 groups of 3).
 - Each group will present a design review of a specific reactor technology or fuel cycle strategy during the final week.
 - **Note:** There is no final exam in this course.

Generative AI Policy

In this course, Generative AI tools (e.g., ChatGPT, Claude, Gemini, Copilot) are treated as **productivity tools**, similar to a calculator or a spell-checker. While AI is unquestionably an aid to productivity, emerging research has shown that it actually can be a detriment to learning if used as a replacement for critical thinking. In this course, we will actively explore how to use these tools effectively and safely, particularly in the Final Project. You are permitted to use these tools, provided you adhere to the following guidelines:

1. **The "Co-Pilot" Rule:** You may use AI to debug code, brainstorm presentation topics, summarize long technical documents, or explain difficult concepts (e.g., "Explain the six-factor formula like I'm a freshman").
2. **The "Lead Engineer" Rule (Final Project):** Your final project is explicitly designed to incorporate AI generation. You are encouraged to use AI to generate outlines, structure arguments, and draft text. However, **you are the Lead Engineer**. You may not submit raw, unverified AI output. You must heavily edit, fact-check, and synthesize the AI's drafts into a rigorous, mathematically sound engineering report.
3. **Verification is Mandatory:** AI models frequently "hallucinate" (invent) facts, citations, and physical constants. In Nuclear Engineering, precision is a safety requirement. **You are 100% responsible for the accuracy of your work.** "The AI told me so" is not a valid defense. Hallucinated references or "context-collapsed" physical parameters in your final project will result in severe grade penalties.
4. **Transparency:** If you use AI significantly for a project (e.g., to write a script that solves the Point Kinetics equations, or to draft sections of your report), you must append a brief note/appendix citing the tool and explaining your methodology and verification process.
5. **Quizzes:** The use of AI tools during quizzes is strictly prohibited.

Course Schedule

		Week	Dates	Topic / Reading
1	1/12, 1/14, 1/16			Context & Foundations L1: The Engineering Context L2: Atomic Structure & Mass-Energy Equivalence L3: The Nuclear Landscape & Excited States
2	(Skip), 1/21, 1/23			Natural Nuclear Engineering I <i>(Monday 1/19: MLK Day - No Class)</i> L4: Neutron Decay & The Physics of Beta Radiation L5: Space Weather & The Atmospheric Source
3	1/26, 1/28, 1/30			Natural Nuclear Engineering II L6: The Carbon-14 Clock & Anthropogenic Effects L7: Alpha Decay & The Radon Hazard L8: Geological Clocks (U-Pb, Rb-Sr) & Earth's Heat
4	2/2, 2/4, 2/6			Neutron Physics L9: Neutron Interactions & Cross Sections L10: The Physics of Neutron Moderation L11: The Physics of Fission (Liquid Drop Model)
5	2/9, 2/11, 2/13			Reactor Theory I: Criticality L12: The Reactor Lattice & Unit Cell Concept L13: The Global Diffusion Equation L14: Geometric Buckling and the Reflected Core
6	2/16, 2/18, 2/20			Reactor Theory II: Kinetics & Feedback L15: The Six-Factor Formula and Prompt Kinetics L16: Delayed Neutrons and the Inhour Equation L17: Reactivity Feedback
7	2/23, 2/25, 2/27			Reactor Theory III: Control L18: Fission Product Poisons (Xe-135 & Sm-149) L19: Control Rod Physics & The S-Curve L20: Flux Shaping & Peaking Factors
8	3/2, 3/4, 3/6			Navy Week: Reflectors & Operations L21: Two-Group Theory & Reflector Savings L22: Guest Lecture: <i>Capt. M. B. Ryan (USN)</i> L23: Historic Lecture: <i>Leighton (Sr) 1961 ASM Address</i> Spring Break (3/9 – 3/13)
9	3/16, 3/18, 3/20			The Road to the Commercial PWR L24: Shippingport & The Seed-Blanket Concept L25: Core Thermal-Hydraulics & Heat Transfer L26: The Commercial PWR
10	3/23, 3/25, 3/27			Alternative Reactor Systems L27: Boiling Water Reactors (BWRs) & The Fukushima Accident

		Week	Dates	Topic / Reading
				L28: Heavy Water Reactors: The CANDU Design & Physics L29: High-Temperature Gas Reactors (HTGR & TRISO Fuel)
11	3/30, 4/1, (Skip)			Fast Reactors & Materials Science L30: Liquid Metal Fast Reactors (LMFR) L31: Radiation Effects on Materials (Embrittlement, Swelling) <i>(Friday 4/3: Good Friday - No Class)</i>
12	(Skip), 4/8, 4/10			The Fuel Cycle <i>(Monday 4/6: Easter Monday - No Class)</i> L32: The Front End (Mining, Enrichment, SWU Calcs) L33: The Back End (Reprocessing, PUREX & Waste/Yucca)
13	4/13, 4/15, 4/17			Radiation Protection & Safety L34: Interaction of Radiation with Matter (Shielding) L35: Biological Dose, Health Physics, and Regulatory Limits L36: Reactor Safety & Licensing
14	4/20, 4/22, 4/24			The Future & Student Presentations I L37: SMRs On Our Horizon L38: The Energy Implications of AI Fri 4/24: Student Design Reviews (Groups 1 & 2)
15	4/27, 4/29			Student Presentations II Mon 4/27: Student Design Reviews (Groups 3 & 4) Wed 4/29: Student Design Reviews (Groups 5 & 6)